

REVIEW ARTICLE

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REVIEW ARTICLE

Modelling of thermal plasmas for arc welding: the role of the shielding gas properties and of metal vapour

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Abstract

The methods used to model thermal plasmas, including treatments of diffusion in arcs in gas mixtures, are reviewed. The influence of thermophysical properties on the parameters of tungsten–inert-gas (TIG) welding arcs, particularly those that affect the weld pool, is investigated using a two-dimensional model in which the arc, anode and cathode are included self-consistently. The effect of changing each of six thermophysical properties on the characteristics of an argon TIG arc is assessed. The influence of the product of specific heat and mass density is found to be particularly important in determining the arc constriction. By examining the influence of the different properties on the heat flux density, current density and shear stress at the anode, it is concluded that the weld pool depth can be increased by using shielding gases with high specific heat, thermal conductivity and viscosity. The effect of metal vapour on the arc and weld pool properties is assessed. The most important effect of the metal vapour is found to be the increased electrical conductivity at low temperatures, which leads to lower heat flux density and current density at the weld pool, implying a shallower weld pool.

(Some figures in this article are in colour only in the electronic version)

1. Introduction

Thermal plasmas are plasmas that are in or close to local thermodynamic equilibrium (LTE). They can be generated by dc or ac arcs, by rf electromagnetic fields or by microwave fields. As in almost all plasmas, it is the electrons that are initially heated. Thermal plasmas are generally at a pressure of 0.1 atm or greater, which leads to a high collision rate and therefore an efficient transfer of energy from the electrons to the heavy species. The heavy species are therefore at or close to the same temperature as the electrons, typically of the order of 1 eV ($\sim 10\,000$ K). This similarity of translational temperatures

for all species is one of the requirements for LTE; others are that the excitation temperatures of all species are equal, as are the chemical reaction temperatures. Typically the conditions for LTE are met in the central region of arc and inductively coupled rf plasmas at 1 atm [1–6], with deviations occurring close to the electrodes [7, 8] and at the edges [9–11].

Thermal plasmas are widely used in industrial applications, including arc welding [12], plasma cutting [13], plasma spraying [14], mineral processing (including plasma remelting and arc furnaces) [15, 16], arc lighting [17], waste treatment [18] and high-power circuit breakers [19, 20]. An emerging application is production of nanoparticles and nanostructures [21, 22].

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Modelling of thermal plasmas is based on the methods of computational fluid dynamics, with modifications to take into account the fact that plasmas are electrically conducting and radiate strongly. A critical factor is the need to use accurate values of the thermodynamic, transport and radiative properties of thermal plasmas. These are calculated using the Chapman–Enskog method of solution of the Boltzmann equation; this is discussed in another paper in this issue [23].

Recent reviews have presented an overview of the state-of-the-art in modelling of thermal plasmas [24, 25]. This includes the treatment of complicated three-dimensional geometries and rapidly varying phenomena, the ability to include deviations from local thermodynamic and chemical equilibrium and treatment of interactions between the plasma and condensed phases. Modelling of thermal plasmas has reached the point where useful predictions can be made that can guide process design, for example in plasma cutting [26], waste destruction [27, 28] and nanoparticle production [29].

In this paper, we review the methods used to model arc plasmas, both in pure gases and in mixtures of gases, paying particular attention to the proper treatment of diffusion when more than one gas is present. We focus on modelling of arc welding, which is the most widespread application of thermal plasmas. The purpose of arc welding is to join metal components; this is achieved by using the intense heat flux from an arc plasma to melt the metal components in the region adjacent to the join. The quality of the weld is determined by many factors, including fluid flow and heat transfer in the weld pool as well as metallurgical factors. Many computational models of arc welding include only the metal in the computational domain, with the arc plasma treated as a heat source at the boundary of the domain [30, 31]. However, since this heat source, and other important parameters at the interface between the plasma and the weld pool, depend strongly on arc current, welding speed, electrode shape, presence of metal vapour and so on, it is not possible to accurately predict the conditions at the interface without modelling the arc plasma.

We examine the influence of the thermophysical properties of the plasma (the thermodynamic properties, the transport coefficients and the net radiative emission coefficient) on the arc parameters, particularly those that affect the heat transfer to and the fluid flow in the molten weld pool. To do this, we present two separate studies. In the first, arcs are modelled in ‘imaginary’ gases, namely gases with the thermophysical properties of argon, but with one of these properties (e.g. thermal conductivity) altered in turn to that of another gas. This allows the influence of each property to be examined separately. This study builds on results previously presented by Tanaka *et al* [12, 32] by examining the influence of all the thermophysical properties and by examining their influence on the weld pool as well as the arc. It complements the recent paper of Murphy *et al* [33] that examined the utility of different gas mixtures in arc welding. In the second study, we examine the influence of metal vapour from the weld pool on the arc properties. We review previous work, highlighting and extending the studies of Yamamoto *et al* [34, 35], which were the first in which the molten weld pool was included self-consistently in the computational domain.

In section 2, we present the equations and methods required to model arc welding; these equations are applicable to all arc plasmas, and we indicate the changes required to treat inductively coupled rf plasmas. In section 3, we apply these equations to modelling of welding arcs in different gases, presenting the study of arcs in ‘imaginary’ gases. In section 4, we investigate the important problem of modelling mixtures of gases, and in particular appropriate treatment of diffusion of the different species. We use the study of the effect of metal vapour in a welding arc to illustrate the discussion. Conclusions are presented in section 5.

2. Modelling of thermal plasmas

2.1. Equations for thermal plasma modelling

Computational modelling of thermal plasmas uses the equations of fluid dynamics, specifically those applicable to a viscous compressible fluid, with modifications to take into account the specific properties of plasmas, in particular their high electrical conductivity and their strong radiative emission. It is usual to assume LTE, which has proved reasonable for the modelling of arc welding plasmas, provided the sheath regions between the arc and electrodes are properly taken into account [12]. A recent review [10] has considered modelling of thermal plasmas out of LTE in detail.

In this section, we will consider an arc plasma containing a single chemical element (or multiple elements if they are assumed to be fully mixed) in LTE. Plasmas in gas mixtures are considered in section 4.

Four conservation equations are required. The equation of mass continuity is

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \underline{v}) = 0, \quad (1)$$

where ρ is the mass density, $\underline{v}$ is the flow velocity and t is time. As in the momentum and energy conservation equations (2) and (4), the two terms on the left-hand side are the time-dependent term and the convective flow term. The equation of momentum conservation is

$$\frac{\partial (\rho \underline{v})}{\partial t} + \nabla \cdot (\rho \underline{v} \underline{v}) = -\nabla P - \nabla \cdot \underline{\underline{\tau}} + \underline{j} \times \underline{B} + \rho \underline{g} \quad (2)$$

where P is pressure, $\underline{\underline{\tau}}$ is the stress tensor, $\underline{j}$ is current density, $\underline{B}$ is magnetic field strength and $\underline{g}$ is acceleration due to gravity. The terms on the right-hand side describe, respectively, the forces due to pressure gradients, viscous stress, the Lorentz or magnetic pinch force and gravity. In Cartesian geometry, the stress tensor is given by

$$\begin{aligned} \tau_{ii} &= \eta \left(2 \frac{\partial v_i}{\partial x_i} - \frac{2}{3} \nabla \cdot \underline{v} \right), \\ \tau_{ij} &= \eta \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right), \quad i \neq j, \end{aligned} \quad (3)$$

where η is the viscosity and v_i and x_i represent the components (v_x, v_y, v_z) and (x, y, z) of the velocity $\underline{v}$ and the displacement

$\underline{r}$, respectively. The equation of energy conservation is

$$\frac{\partial(\rho h)}{\partial t} + \nabla \cdot (\rho \underline{v} h) = \frac{j^2}{\sigma} - U - \nabla \cdot \left(\frac{k}{c_p} \nabla h \right) + \frac{5k_B}{2ec_p} \underline{j} \cdot \nabla h, \quad (4)$$

where h is the enthalpy, σ is the electrical conductivity, U is the net radiative emission coefficient, k is the thermal conductivity, c_p is the specific heat at constant pressure, k_B is Boltzmann's constant and e is the electronic charge. The terms on the right-hand side describe, respectively, resistive heating, radiative emission, thermal conduction and energy transfer arising from the flow of electrons. The enthalpy is the integral of specific heat with respect to temperature, and the temperature at any position is easily derived from the enthalpy at that position. Finally, the equation of current continuity is

$$\nabla \cdot (\sigma \nabla \phi) = 0, \quad (5)$$

where ϕ is the electric potential (so $\underline{j} = -\sigma \nabla \phi$).

The magnetic field strength $\underline{B}$, which appears in (2), also has to be calculated. This can be done by solving for the magnetic potential $\underline{A}$:

$$\nabla^2 \underline{A} = -\mu_0 \underline{j}, \quad (6)$$

and using $\underline{B} = \nabla \times \underline{A}$. For axisymmetric geometry, it is usually a good approximation to use only the azimuthal component of the magnetic field strength

$$B_\theta(r) = \frac{\mu_0}{r} \int_0^r r' j_z dr', \quad (7)$$

which is derived from $\nabla \times \underline{B} = \mu_0 \underline{j}$, with only the axial component of the current density j_z considered.

The above set of equations is valid for laminar flow, which prevails in most welding arcs. If the flow is turbulent, additional equations are required. The standard treatment is the $K - \varepsilon$ method, which requires solution of partial differential equations for turbulent kinetic energy and its rate of dissipation. From these, turbulent contributions to the viscosity and thermal conductivity are derived [36, 37]. More sophisticated treatments have also been used [24].

For an rf inductively coupled plasma, the current density $\underline{j}$ in equation (6) is the sum of $\underline{j}_{\text{coil}}$, the current density induced in the induction coil by the potential across the ends of the coil, and $\underline{j}_{\text{ind}}$, the current density developed in the plasma and the coil by the induced electric field [38]. The magnetic field intensity and electric field are calculated using standard equations: $\mu_0 \underline{H} = \nabla \times \underline{A}$ and $\underline{E} = -\partial \underline{A} / \partial t$. The Lorentz force term in equation (2) is replaced by $\frac{1}{2} \mu_0 \text{Re}(\underline{E} \times \underline{H}^*)$ and the resistive heating term in equation (4) by $\frac{1}{2} \sigma (\underline{E} \cdot \underline{E}^*)$.

Under the assumptions of LTE and a single gas or fully mixed gases, the thermophysical properties (the thermodynamic properties ρ , c_p and h , the transport coefficients η , k and σ and the net radiative emission coefficient U) depend only on the local temperature and pressure. In most cases the pressure can be assumed constant, so the thermophysical properties are then a function of temperature only. Details of the methods used in their calculation have been given by Boulos *et al* [39]. The thermodynamic and transport properties used in the work

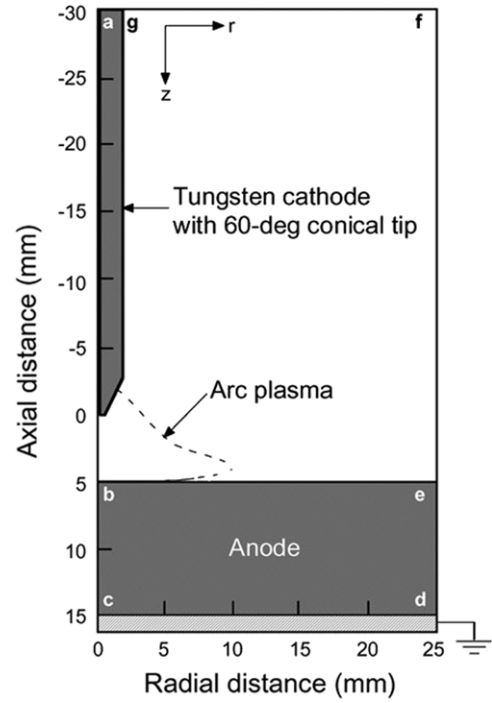


Figure 1. Computational domain for TIG welding, showing the external and internal boundaries. From [12].

presented here were calculated using the methods given previously by Murphy and Arundell [40] for argon and nitrogen plasmas, Murphy [41] for helium plasmas and Murphy [42] for hydrogen plasmas. Properties for the argon–iron vapour and helium–iron vapour mixtures were calculated using methods similar to those given in [43] for argon–copper vapour mixtures. Net radiative emission coefficients were determined using the method of Cram [44], assuming a 1 mm absorption length, which is appropriate for the 5 mm long welding arc considered here. Net radiative emission coefficients for the mixtures were calculated using linear interpolations based on mole fractions, as recommended by Cressault *et al* [45]. The importance of each of these thermophysical properties in determining the properties of the arc plasma and its influence on the weld pool is the subject of section 3.

2.2. Boundary conditions for modelling of arc welding

The computational domain for the case of tungsten–inert-gas (TIG) welding is shown in figure 1. Two-dimensional axisymmetric geometry is used. In the usual TIG welding configuration, the upper electrode is the cathode, and is made of tungsten, typically doped with a rare-earth oxide, such as thorium, to reduce the work function. The electrode has a conical tip and remains solid. The lower electrode, also known as the workpiece, is the anode; the molten region of the workpiece is the weld pool. In metal–inert-gas (MIG) welding, the upper electrode is the anode, and is a wire composed of the same or a similar metal to the workpiece. The wire melts, forming droplets of liquid metal that fall to the workpiece.

The boundary conditions at the external boundaries were given in [12] and are repeated briefly here. On the axis ($r = 0$), derivatives of all quantities with respect to radius are set to zero.

The temperature at the external boundaries is set to 300 K. The axial velocity boundary condition at the upper boundary gf in figure 1 is used to define the input flow rate of the shielding gas. The derivatives of the velocity components are zero at the radial boundary fe. The pressure needs only to be defined at one point; it is set to 1 atm at point f. The axial current density at ag is chosen so that the integral over the cross-section at the top of the cathode is equal to the arc current. The electrical potential is set to 0 V at point d.

The internal boundaries, at the arc–anode and arc–cathode interfaces, have been considered in detail for TIG welding in [33]. Here we present a summary. We have to consider energy transfer at both boundaries and momentum transfer at the arc–anode boundary. Since the cathode remains solid in TIG welding, no momentum transfer occurs at the arc–cathode boundary and no mass transfer occurs at either boundary. Note that in MIG welding, since both electrodes melt, momentum and mass transfer have to be considered at both boundaries. Also, if vaporization of the electrodes is taken into account, this implies mass transfer occurs; however, as discussed in section 4.2, this is usually treated by a boundary condition for vapour concentration at the electrode surface.

Heat transfer occurs by thermal conduction, and also through the transfer of charged species across the arc–electrode interfaces. Blackbody radiation from the electrodes is small compared with the other terms, and can be neglected. The net heat flux to the cathode surface is given by

$$S_c = -|j_e|\phi_{wc} + |j_i|V_i - k \frac{\partial T}{\partial n}, \quad (8)$$

where the terms describe thermionic cooling, heating due to the ion current and thermal conduction respectively [46]. The thermionic cooling is the product of the work function of the cathode material ϕ_{wc} and the electron current density j_e and the ion heating is the product of the ion current j_i and the ionization energy of the shielding gas V_i . In the thermal conduction term, n denotes the displacement normal to the electrode surface.

Treatment of heat transfer at the arc–anode interface is complicated by the fact that the temperature adjacent to the anode is relatively low, so that the electron density and thus electrical conductivity are negligible if LTE is assumed. This can be overcome by taking into account diffusion of electrons into this region, which requires a separate continuity equation for electrons [47]. A simpler approach is the LTE-diffusion approximation [48], in which the mesh size adjacent to the anode is chosen as approximately equal to the electron diffusion distance. This means that there are no calculations made at points at which the electron density is zero. The method has been shown to predict heat flux and current densities that are in agreement with more complicated models and with experiment. The heat flux at the surface of the anode is given by

$$S_a = j_e\phi_{wa} - k \frac{\partial T}{\partial z}, \quad (9)$$

where ϕ_{wa} is the work function of the anode material. The terms describe heating by electron capture and thermal conduction, respectively.

In TIG welding, the molten weld pool forms part of the anode, and boundary conditions for momentum transfer across

the arc–anode interface have to be considered for this region. These are, respectively,

$$\tau_a - \tau_w + \frac{d\gamma}{dT} \frac{\partial T}{\partial r} = 0 \quad (10)$$

parallel to the anode surface and

$$P_a - P_w + \gamma\kappa = 0 \quad (11)$$

perpendicular to the anode surface. The terms in (10), which is given in a form appropriate to axisymmetric geometry are, respectively, the shear stresses on the arc side and the weld pool side of the interface, and the Marangoni term, which describes the radial variation of the surface tension γ due to the radial gradient of temperature close to the weld pool surface. The factor $d\gamma/dT$ is negative for most pure metals, but impurities such as oxygen and sulfur can lead to positive values. The terms in (11) are, respectively, the pressures on the arc side and the weld pool side of the interface, and the product of the surface tension and the curvature κ of the interface.

Our focus here is the influence of the arc plasma on the weld pool flow and shape. The important factors are (a) the heat flux to the anode, given by (9); (b) the current density at the anode surface, which features in the first term in (9) and is also significant in determining the flow in the weld pool through the magnetic pinch ($\vec{j} \times \vec{B}$) force [12]; (c) the shear stress due to the arc τ_a , which is the first term in (10) and (d) the arc pressure, which is the first term in (11). The other factors that determine the weld pool flow and shape are properties of the liquid metal; of these the most important is the Marangoni term [12].

2.3. Computational details

Equations (1) to (5) and (7) are solved by the finite-volume method of Patankar [49], using the internal boundary conditions (8) to (11) and the external boundary conditions presented in section 2.2. The calculations presented in section 3 are steady state, so the first term in equations (1), (2) and (4) was not required in this case. The SIMPLER algorithm used by Patankar to determine the coupling between the velocity and pressure fields was replaced by the SIMPLEC algorithm of van Doormaal and Raithby [50]. The computational domain was shown in figure 1. We used a variable mesh size, with about 80 mesh points in both the radial and the axial directions. The finest mesh (about 0.01 mm) was used at the conical tip of the cathode. The axial mesh size at the surface of the anode was 0.5 mm; this is chosen to approximate the diffusion length of electrons near the anode, as described in [48].

Detailed comparisons of the predictions of the model with experiment have been performed. For example, the model has been benchmarked against measurements of arc temperature distribution and of weld depth and shape for the case of argon shielding gas and a stainless steel workpiece [12, 51], and against measurements of weld depth in the case of helium shielding gas [12]. For the case of metal vapour production in argon and helium arcs, detailed benchmarking has not been performed; however, observation of the condensed metal vapour on the anode has been used to confirm the predicted

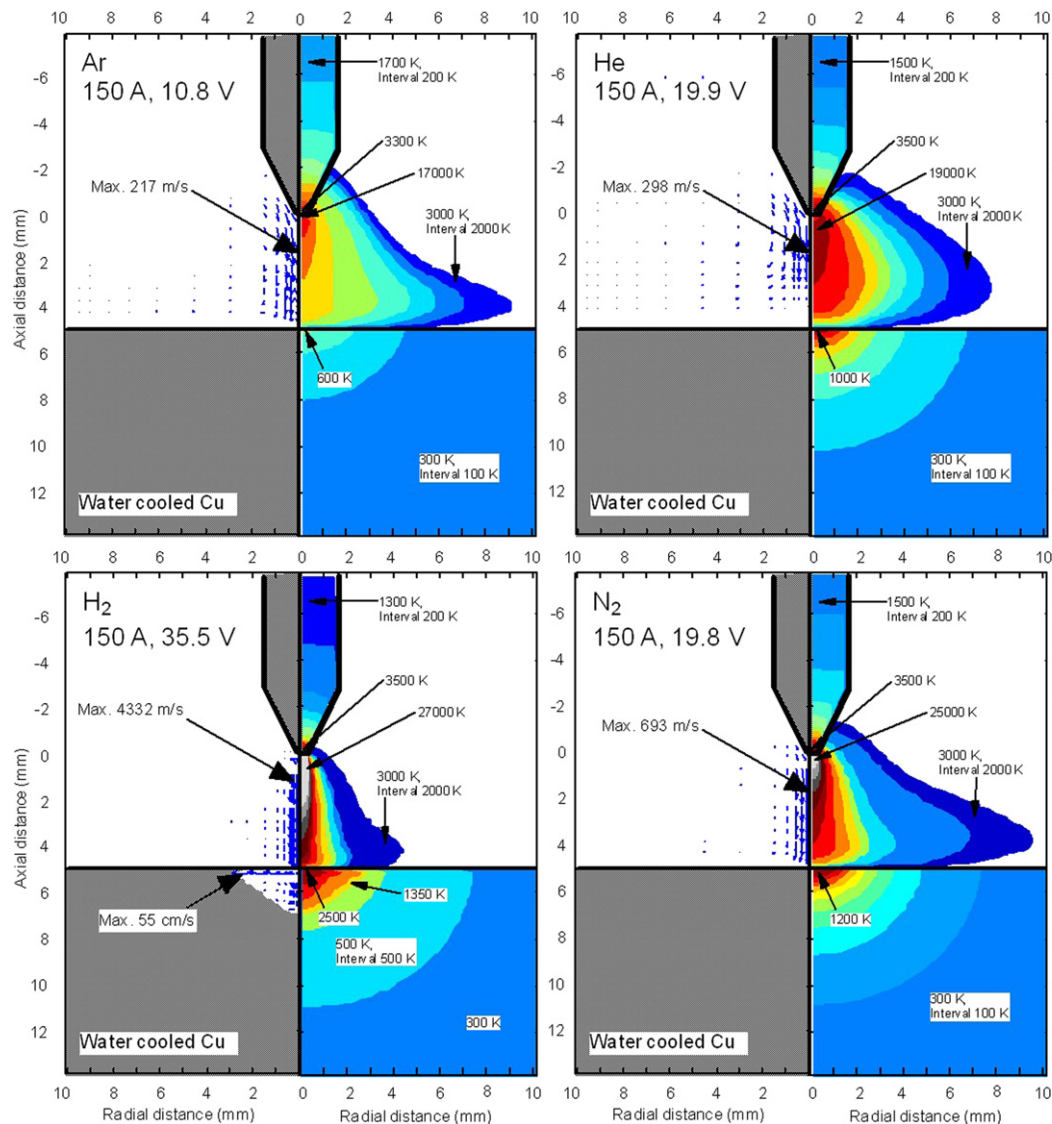


Figure 2. Temperature fields and velocity vectors in the arc and electrodes for 150 A arcs in argon, helium, hydrogen and nitrogen at 1 atm, with a water-cooled copper cathode. The maximum temperatures in the arc, cathode and anode are given. The temperature interval is 2000 K in the arc, 200 K in the cathode, and 100 K in the anode (500 K for hydrogen). The maximum velocity is given in the arc, and in the anode in the case of hydrogen. The arc voltage is given in each case. After [32].

strong increase in metal vapour for a helium arc compared with an argon arc [34].

We used the following standard conditions for our TIG arc welding calculations: an arc current of 150 A, an arc length of 5 mm, a gas flow rate of 10 L min^{-1} and atmospheric pressure. The cathode was of thoriated tungsten, with 3.2 mm diameter and a 60° conical tip angle, with the tip flattened to give a radius of curvature of 0.1 mm. For the calculations presented in section 3, a water-cooled copper anode was used; this is because the focus of this section is on the arc properties, and not of those of a molten weld pool. In section 4, we used a 304 stainless steel anode, so as to allow the effects of melting and vaporization of the anode material to be taken into account.

3. Modelling of arcs in imaginary gases

The properties of a TIG arc are strongly dependent on the shielding gas. This is clearly apparent in figure 2, which

shows the temperature and flow fields for arcs in argon, helium, hydrogen and nitrogen. The argon arc has the lowest voltage, and the lowest maximum arc temperature, anode temperature and gas flow velocity. The molecular gas arcs, particularly the hydrogen arc, are more constricted, in the sense that the area of the cross-section for which temperatures are above a given value is smaller. The heat flux to the anode is particularly large for the hydrogen arc, resulting in the formation of a molten weld pool even for a water-cooled copper anode. The helium arc does not have the bell shape typical of most TIG arcs. Further, while the gas flow velocity in the helium arc is higher than that of the argon arc close to the cathode, it is lower near the anode.

The properties of the arcs in different gases are determined by the different thermophysical properties of the gases, which appear in the conservation equations. These are the mass density, specific heat at constant pressure and specific enthalpy (the thermodynamic properties), viscosity,

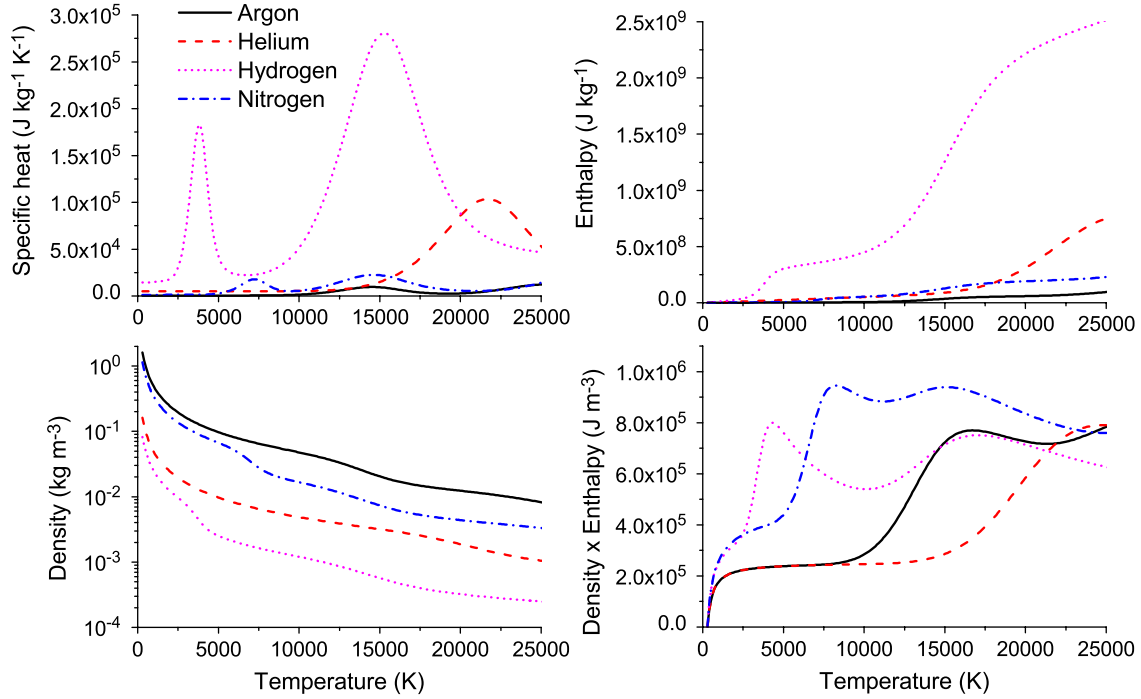


Figure 3. Dependence of specific heat at constant pressure, specific enthalpy, mass density and the product of mass density and specific enthalpy on temperature for argon, helium, hydrogen and nitrogen at 1 atm.

thermal conductivity and electrical conductivity (the transport coefficients) and the net radiative emission coefficient. Figure 3 shows the thermodynamic properties for the four shielding gases considered here. The product of the specific enthalpy and mass density is also shown; this will be used later.

The mass density is determined largely by the atomic or molecular weight of the gas, although it is reduced upon dissociation and ionization (for example, dissociation of a diatomic molecular gas leads to a halving of the density according to the ideal gas law, since the pressure is constant). As expected, argon is the densest gas, and hydrogen and helium the least dense.

The specific heat and enthalpy are related by

$$c_p = \left. \frac{dh}{dT} \right|_p. \quad (12)$$

All gases have peaks in their specific heat at the temperatures of ionization (around 15 000 K for argon, hydrogen and nitrogen and around 22 000 K for helium, for the first ionization). Molecular gases have an additional peak at the temperature of dissociation (around 3500 K for hydrogen and 7000 K for nitrogen). There are corresponding increases in the enthalpy, as expected from (12). Hydrogen has the highest specific heat and enthalpy and argon the lowest. This is mainly because these quantities are defined on a per unit mass basis. When the enthalpy and mass density are multiplied, the mass dependence of each quantity cancels, and the differences between the gases are smaller. The product is largest for nitrogen and hydrogen, while the product is almost identical up to about 10 000 K for argon and helium.

Figure 4 shows the transport coefficients and the net radiative emission coefficient for the four gases. The thermal conductivity is made up of components due to the translational

motion of heavy species, the internal energy of heavy species, the translational motion of electrons and the reaction thermal conductivity [39, 40]. The component due to the translational motion of heavy species is generally the largest at low temperatures, and is approximately inversely proportional to the collision cross-sections between the species present at the relevant temperature, and to the square root of the atomic mass. Hydrogen has the lowest mass, and the collision cross-section for interactions between neutral species is lowest for helium, so the heavy species translational component is largest for these gases. The electron translational thermal conductivity increases with temperature, and is approximately the same for all the gases. The reaction thermal conductivity has peaks at the same temperatures as the specific heat; these are associated with the transport of energy required for dissociation and ionization reactions. This leads to large peaks at the dissociation temperatures of hydrogen and nitrogen, and peaks at the ionization temperatures of each gas. The internal energy component of the thermal conductivity is relatively small.

Viscosity is proportional to the square root of the atomic mass and inversely proportional to the collision cross-sections of the species present. Hydrogen has the lowest viscosity because of its low mass. The weak collision cross-section for interactions between helium atoms means that it has a relatively high viscosity, despite its low mass. Further, since helium ionizes at high temperature, the strong Coulomb interaction does not dominate the collision cross-sections until temperatures of around 20 000 K, meaning that the viscosity of helium remains high until about this temperature.

The electrical conductivity increases with the electron density. Since argon, nitrogen and hydrogen ionize at about the same temperature, they have similar electron densities and therefore electrical conductivities. The electrical conductivity

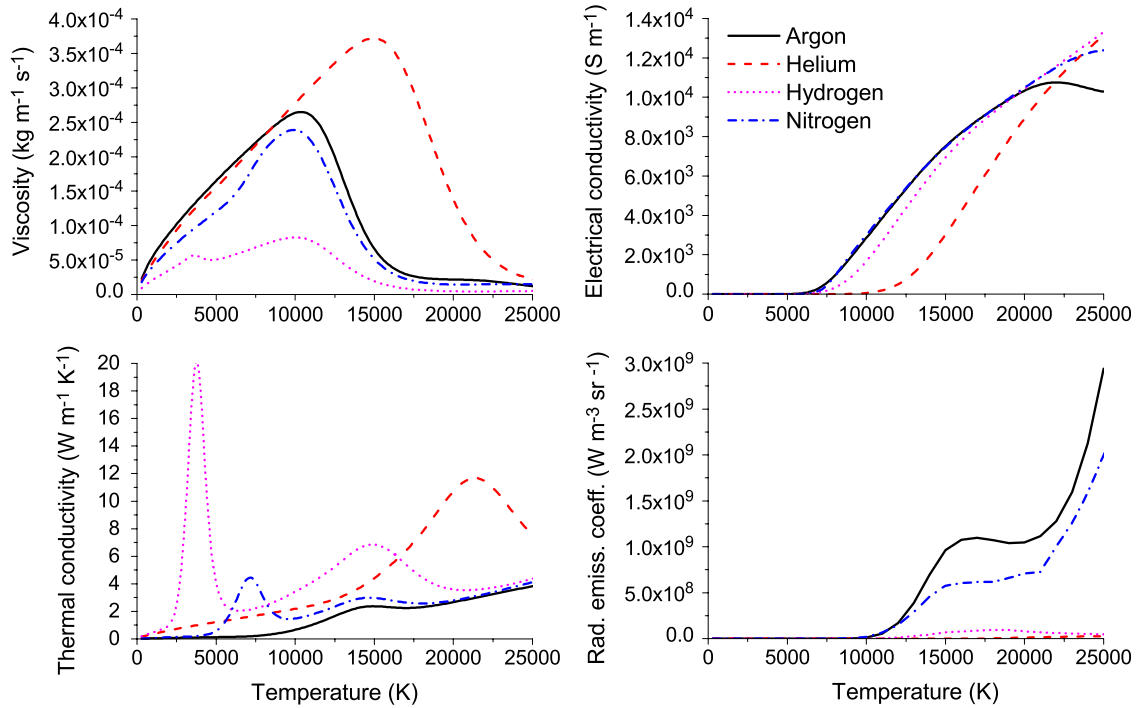


Figure 4. Dependence of viscosity, electrical conductivity, thermal conductivity and net radiative emission coefficient on temperature for argon, helium, hydrogen and nitrogen at 1 atm.

of helium is lower than that of the other gases until its ionization temperature is reached.

The net radiative emission coefficient is largest for argon and nitrogen; hydrogen and helium radiate much more weakly because they have only one and two electrons, respectively.

Since most of the thermophysical properties of a given shielding gas differ greatly from those of all the other gases, it is difficult to isolate those particular properties that are responsible for the different behaviours of arcs in different gases. However, computational modelling is not restricted to real gases such as argon and helium. It is possible to construct ‘imaginary gases’ by replacing one thermophysical property by that of another gas. For example, the electrical conductivity of argon can be replaced by that of helium, while retaining all the other properties of argon, to give ‘low-conductivity argon’. By simulating arcs in a range of such imaginary gases, it is possible to vary one thermophysical property at a time, and therefore to clarify the influence of each property on arc behaviour. A similar approach has been previously used in an investigation of atmospheric-pressure inductively coupled plasmas [52].

We chose to use argon thermophysical properties, replacing in turn the thermal conductivity, electrical conductivity, specific heat, density, viscosity and net radiative emission coefficient by those of helium. In all cases, the helium property is significantly different from that of argon. In the case of the specific heat, we used a composite specific heat, as shown in figure 5: that of helium for temperatures below 8900 K and that of carbon dioxide above 8900 K; this ensures the specific heat is higher than that of argon for the full range of temperatures present in the arc. The enthalpy was also changed accordingly, because the enthalpy is related to the specific heat by (12).

The temperature and velocity distributions in the arcs in the six imaginary gases are shown in figure 6. We first consider the effect of increasing the thermal conductivity to that of helium, shown in figure 6(a). It has been previously argued [53–56] that increasing the thermal conductivity of the plasma will increase the constriction of an arc; the mechanisms that have been suggested are through an increase in the current density near the arc axis [53], greater heat removal from the arc core [54], or to maintain the thermal balance of the arc [55]. However, as shown by comparing figure 6(a) with the pure argon arc shown in figure 2, increasing the thermal conductivity to that of helium leads to a less constricted arc, with a lower temperature on axis than an argon arc. The current density on axis is also lower near the cathode, as indicated by the lower axial flow velocity (maximum value is 173 m s^{-1} , compared with 217 m s^{-1} for the argon arc); the flow is driven by the magnetic pinch ($\vec{j} \times \vec{B}$) term in the momentum conservation equation (2).

The changes can be understood as follows: increasing the thermal conductivity increases the rate of heat transfer from the high-temperature central region to the arc fringes, leading to a less peaked temperature profile. Since electrical conductivity increases with temperature (see figure 4), this in turn leads to a less peaked current density profile, further contributing to the flatter temperature profile. Despite these changes, the anode temperature is higher than for the argon arc, reaching 900 K; this will be considered later.

Note that in the case of molecular gases, the large peaks in the thermal conductivity at low temperatures (see figure 4) may lead to cooling of the outer regions, thereby forcing the current to flow preferentially in the centre of the arc.

Figure 6(b) shows that a highly constricted arc is obtained when the specific heat and enthalpy are increased to those

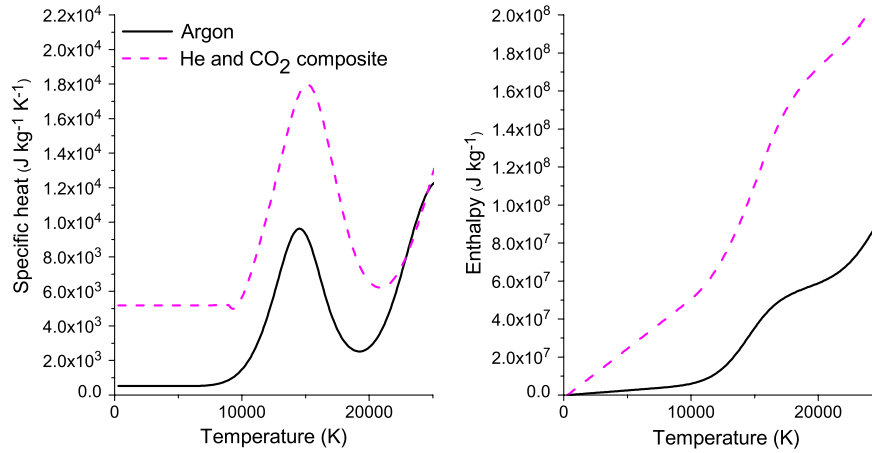


Figure 5. Specific heat and enthalpy of argon and the composite gas (helium for temperatures up to 8900 K and carbon dioxide for temperatures above 8900 K).

shown in figure 5. The mechanism by which this constriction occurs was discussed by Tanaka and Lowke [12]. They argued that at any axial position z , the total power flowing towards the anode can be approximated by the product of the arc current I and the potential difference V_z between the cathode and the plasma at the axial position. This neglects the relatively small radiation and cathode conduction losses. Using

$$IV_z = \overline{\rho_z} \overline{h_z} \overline{v_z} A_z, \quad (13)$$

where $\overline{\rho_z}$, $\overline{h_z}$ and $\overline{v_z}$ are, respectively, mass density, enthalpy and velocity averaged over the arc cross-sectional area A_z at axial position z , it is clear that for the same values of I and V_z , an increase in the value of the enthalpy (or equivalently the specific heat) will lead to a decrease in A_z , and hence a more constricted arc. This effect was termed the *thermal pinch effect* [12], although it should be noted that this term has been used in other contexts, for example to describe the influence of cold gas in constricting a plasma torch plume. In comparing the high specific heat arc shown in figure 6(b) with the argon arc shown in figure 2, we note that the total voltage, and therefore IV_z , has approximately doubled. However, the enthalpy, as shown in figure 5, is ten times greater than that of argon up to 10 000 K, and at least three times greater up to 20 000 K. Further, the flow velocity is greatly increased. The net result is a large decrease in the arc cross-sectional area, according to (13).

Constriction of the arc leads to increased arc temperature and axial flow velocity. These are a consequence of the increased current density near the arc axis associated with the arc constriction. The arc temperature is higher because the ohmic heating (the j^2/σ term in the energy conservation equation (4)) is stronger. The flow velocity is increased by the larger magnetic pinch force ($j \times B$) term in the momentum conservation equation (2).

Figure 6(c) shows the effect of decreasing the mass density to that of helium. As expected from (13), this leads to an increase in the cross-sectional area A_z ; i.e. it has the opposite effect to increasing the specific heat and enthalpy. For a helium arc, figure 3 shows that the decrease in mass density and increase in enthalpy relative to an argon arc approximately

cancel over much of the temperature range. For this reason, the helium arc is not greatly constricted compared with the argon arc, as was shown in figure 2. The constriction of the hydrogen and nitrogen arc is much greater, since the product of mass density and enthalpy is much larger for these gases.

The effect of a reduction in electrical conductivity is shown in figure 6(d). We see from figure 4 that the electrical conductivity of helium is close to zero for temperatures below 10 000 K, and more than a factor of 2 less than that of argon for temperatures up to about 16 000 K. This means that current only flows in the high-temperature central regions of the arc, thereby increasing the current density near the arc axis. The effect is not as strong as that of increasing the specific heat and enthalpy. Nevertheless, it is probably the most important factor in the greater constriction observed for the helium arc relative to the argon arc.

Figure 6(e) shows the effect of increasing the viscosity to that of helium. The main effect is on the axial flow velocity, which is reduced near the arc axis, but increased at radii greater than 1 or 2 mm. This flattening of the radial profile of axial velocity is a direct consequence of the increased viscosity, which determines the transfer of momentum in a direction transverse to the flow. The influence on the arc temperature is small. Hydrogen has a much lower viscosity than the other gases, with the consequence that the flow velocity retains a strongly peaked radial profile at all axial positions (see figure 2).

Finally, we show in figure 6(f) the effect of a reduction in the net radiative emission coefficient. As expected from a reduction in this energy loss term, the main effect is an increase in the arc temperature. There is no obvious change in arc constriction, and the anode temperature is unchanged. The effect is relatively small, because for arcs with these relatively low currents, only a small volume of plasma is at temperatures above 20 000 K, for which the radiative emission coefficient begins to increase steeply for argon (see figure 4) and for which radiation therefore starts to dominate the energy balance.

The influence of the different thermophysical parameters on the weld pool in TIG welding can be assessed by calculating the heat flux density, current density, arc pressure and shear

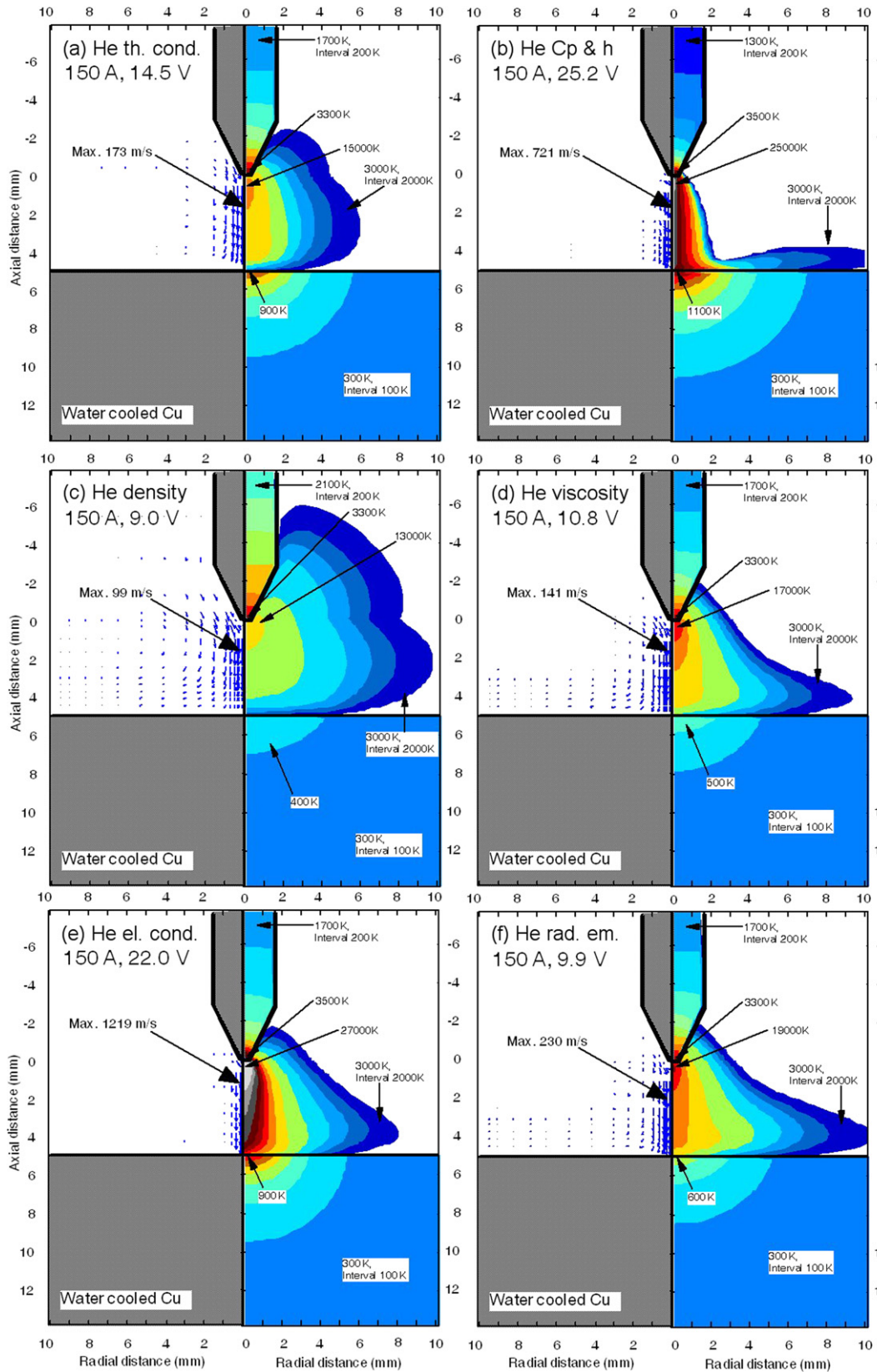


Figure 6. Temperature fields and velocity vectors in the arc and electrodes for 150 A arcs in imaginary gases at 1 atm. The imaginary gases are argon with (a) thermal conductivity, (b) specific heat and enthalpy, (c) mass density, (d) viscosity, (e) electrical conductivity and (f) net radiative emission coefficient, altered to that of helium. In the case of (b) the composite specific heat shown in figure 5 is used, with enthalpy calculated from this. Contours and labelling are as per figure 2.

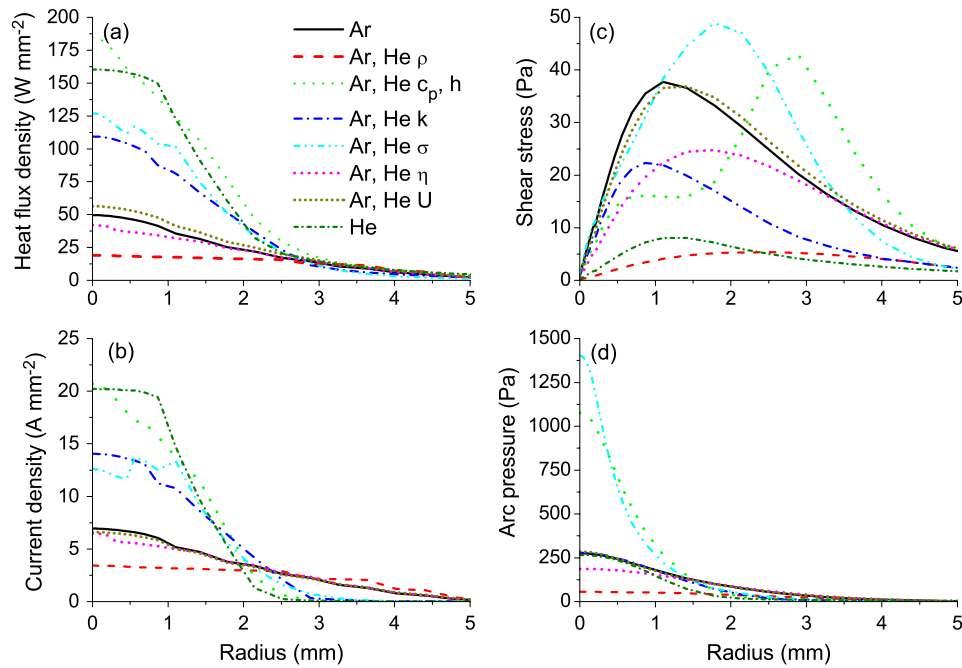


Figure 7. Radial dependence of (a) heat flux density, (b) current density, (c) shear stress and (d) arc pressure, all at the anode, for arcs in argon, helium, and in the imaginary gases: argon with helium mass density, argon with helium/carbon dioxide specific heat and enthalpy, argon with helium thermal conductivity, argon with helium electrical conductivity, argon with helium viscosity, and argon with helium net radiative emission coefficient.

stress in the arc immediately above the anode. As discussed in section 2.2, a deep weld pool requires a high heat flux density to increase melting, a high current density to drive liquid metal flow downwards through the magnetic pinch force and a low shear stress to minimize the flow radially outwards. The radial distribution of these properties for the different imaginary gases is shown in figure 7.

The heat flux density, calculated using (9), is shown in figure 7(a). The heat flux density on axis is increased by more than a factor of 3 relative to argon for the imaginary gas with high specific heat and enthalpy, and increased by more than a factor of 2 for the imaginary gases with high thermal conductivity and low electrical conductivity. A strong decrease in the heat flux density on axis is found for the imaginary gas with low mass density. As expected, the same trends are found for the maximum anode temperatures given in figure 6.

The two components of the heat flux density are the electron condensation term, which is proportional to the current density, and the thermal conduction term. The current density is shown in figure 7(b). The same trends are found as for the heat flux density, indicating that changes in the current density are at least partly responsible for the changes in the heat flux density. The changes in current density for the different imaginary gases are a consequence of the strong dependence of electrical conductivity on temperature, shown in figure 4. For a given electric field, more current will flow in a region at higher temperature, which in turn leads to further ohmic heating. The case of the imaginary gas with low thermal conductivity is interesting, since the current density was found to be lower than that of the argon arc near the cathode. Near the anode, however, the current density is higher than that of the argon arc. This is a consequence of the temperature profile near the anode;

the region for which the temperature is larger than 7000 K (the temperature range for which the electrical conductivity is at least 10% of its maximum value) is far smaller than for the argon arc, and the current is thus restricted to flowing through a smaller region.

The arc pressure at the anode is determined by the axial flow velocity near the anode, while the shear stress is determined by the radial flow velocity and the viscosity. The radial flow velocity is linked to the axial flow velocity by mass continuity. The arc pressure on axis (figure 7(d)) is largest for the imaginary gases with high specific heat and enthalpy, and low electrical conductivity, and lowest for the imaginary gas with low density. There is a strong correlation between the arc pressure on axis and the maximum flow velocity in the arc, shown in figure 6. The exception is the imaginary gas with high viscosity; the increased viscosity leads to stronger radial transfer of momentum as the flow approaches the anode, and therefore a lower axial velocity near the anode.

Similar trends are found for the shear stress near the anode (figure 7(c)). It is interesting that the peak in the shear stress for the imaginary gas with high specific heat and enthalpy occurs at a larger radius than the other gases; this is because the temperature near the anode is very high (above 15 000 K) at low radii, so that the viscosity is low at these radii (see figure 4). The shear stress for the imaginary gas with high thermal conductivity is also lower than that of argon, even though the arc pressure is very similar. This is also a consequence of the relatively low viscosity near the anode, in this case because the arc temperature near the anode is below the 10 000 K for which viscosity reaches its maximum.

To maximize the depth of the weld pool requires, as noted above, high heat flux and current density, and low shear stress,

at the anode. Based on the results shown in figure 7, high heat flux and current density require a gas with high specific heat and enthalpy, high density, low electrical conductivity and high thermal conductivity. Low shear stress requires a gas with low density, high viscosity, high thermal conductivity, low electrical conductivity, and since the flow near the axis of the weld pool is most important, high specific heat and enthalpy. These requirements are to some extent contradictory; however, high thermal conductivity and high specific heat and enthalpy satisfy both requirements, and high viscosity favours low shear stress while having little effect on the heat flux and current density. Of the real gases considered in figure 3, hydrogen and nitrogen have the properties closest to those required to give high heat flux and current density at the anode, since they have the highest products of specific heat and density and high thermal conductivity. Helium has a much lower electrical conductivity than the other gases up to 20 000 K, and high thermal conductivity. However, it has the approximately the same product of specific heat and density as argon. Helium has the properties, apart from its low electrical conductivity, required to minimize shear stress at the anode: low density, high viscosity and high specific heat. Some of the properties of nitrogen and hydrogen favour lower shear stress, while others promote higher shear stress.

Calculations indicate that, compared with argon arcs, the heat flux to the anode and the current density at the anode are larger in helium, hydrogen and nitrogen arcs, and that the shear stress at the anode is greater in hydrogen and nitrogen arcs, but smaller in helium arcs [32, 33]. It is found in practice that arcs in hydrogen, nitrogen and helium all give deeper weld pools than argon arcs. Pure hydrogen and nitrogen are not suitable shielding gases because they form unstable arcs; however, mixtures of just a few per cent of these gases with argon can give deeper weld pools than pure argon [33]. Knowledge of the importance of the different thermophysical properties on arc parameters has allowed the design of new shielding gas mixtures; for example ternary gas mixtures recently introduced by Taiyo Nippon Sanso Corporation [57], and the addition of hydrogen to carbon dioxide to give deeper weld pools [58]. Ternary gas mixtures have long been used in plasma spraying. For example, mixtures of argon, helium and hydrogen are used to take advantage of the high thermal conductivity of hydrogen and the high viscosity of helium; the former increases the heat flux to the sprayed particles and the latter delays entrainment of the surrounding air [59–61].

4. Modelling of the transport and influence of metal vapour in welding arcs

The presence of metal vapour in the arc is inevitable in arc welding because of the high temperature of the molten metal in the weld pool. In MIG welding, metal vapour also emanates from the molten wire electrode and the droplets. The transport of metal vapour in the arc plasma is an important subject for investigation, since the vapour changes the arc properties, and therefore the properties of the weld pool. Further, the vapour is a precursor in the formation of welding fume, which is a recognized occupational health hazard for welders [62].

In this section, we examine the modelling of thermal plasmas containing more than one chemical element, paying particular attention to the treatment of diffusion of the elements with respect to each other. We then present results of calculations of iron vapour concentration in argon and helium arcs, focussing on the effect of iron vapour on the thermophysical properties of arcs in argon and helium and the consequent changes to the heat flux and current density to the weld pool.

4.1. Modelling of diffusion in thermal plasmas

In the most general treatment of plasmas containing more than one chemical element, mass conservation equations for individual species (e.g. Ar, Ar⁺, Fe, Fe⁺, e[−])

$$\frac{\partial \rho Y_i}{\partial t} + \nabla \cdot (\rho \mathbf{v} Y_i) + \nabla \cdot \mathbf{J}_i = r_i \quad (14)$$

have to be solved everywhere in the plasma. Here Y_i is the mass fraction of species i and r_i is the net rate of production of species i due to chemical reactions. For a gas or plasma containing N species, the diffusion mass flux of species i , $\mathbf{J}_i$, is given by

$$\mathbf{J}_i \equiv m_i n_i \mathbf{v}_i = \frac{m_i n^2}{\rho} \sum_{j=1}^N m_j D_{ij} \mathbf{d}_j - D_i^T \nabla \ln T, \quad (15)$$

where $\mathbf{v}_i$ is the diffusion velocity of species i (relative to the mass-average velocity) and m_i and n_i are, respectively, the mass and the number density of the i th species [63]. The ordinary diffusion coefficients D_{ij} and thermal diffusion coefficients D_i^T are in this case multicomponent diffusion coefficients. Their calculation requires values of the mole fractions and the masses of all the species present and the collision cross-sections for binary interactions between each pair of species present. The driving force term $\mathbf{d}_j$ is given by

$$\mathbf{d}_j = \nabla x_j + \left(x_j - \frac{n_j m_j}{\rho} \right) \nabla \ln P - \frac{n_j m_j}{P \rho} \left(\frac{\rho}{m_j} \mathbf{F}_j - \sum_{l=1}^N n_l \mathbf{F}_l \right). \quad (16)$$

The three terms describe, respectively, diffusion due to gradients in the mole fraction x_j , the pressure P and the external forces $\mathbf{F}_j$ acting on species j . Because diffusion velocities are defined with respect to the mass-average velocity, only $N - 1$ need be calculated, and (14) has to be solved for only $N - 1$ species.

Equations (14) to (16) are valid for neutral gases and plasmas. However, in the case of a plasma, the calculation of the diffusion coefficients has to take into account ambipolar diffusion, which arises because electrons diffuse more rapidly than ions because of their lower mass. The consequent charge separation induces an electric field, known as the ambipolar electric field, which slows the diffusion of electrons and increases the diffusion velocity of ions. To take this into account, the external force term in (16) is written

$$\mathbf{F}_i = e Z_i (\mathbf{E} + \mathbf{E}^a), \quad (17)$$

thereby taking into account both an externally applied electric field $\vec{E}$ and the ambipolar electric field $\vec{E}^a$, where Z_i is the charge number of species i . The ambipolar electric field can be treated either explicitly, by using (17) in (16), or implicitly, by modifying the expressions for the multicomponent diffusion coefficients [64, 65]. In the latter case, only the externally applied electric field $\vec{E}$ is retained in equation (17).

Since calculations of diffusion require the determination of an ordinary diffusion coefficient for each pair of species and thermal diffusion coefficients for each species, they are computationally expensive. (The situation is actually not quite as bad as it may seem, since only $N(N-1)/2$ of the ordinary and $N-1$ of the thermal diffusion coefficients are linearly independent.) As a consequence, simplified methods are generally used. There are a range of methods of varying levels of simplicity and accuracy [43, 67]. They can be divided into two classes: those that are applicable only to plasmas in LTE and those that are also applicable for plasmas out of LTE. We consider each class in turn.

For a plasma in LTE, it is possible to group the different species into ‘gases’. (In fact this is possible for plasmas in local chemical equilibrium, a less restrictive requirement [10], but here we will use LTE.) For example, in an argon plasma containing iron vapour, the species Ar, Ar⁺, Ar²⁺, Ar³⁺, etc are grouped as argon and the species Fe, Fe⁺, Fe²⁺, Fe³⁺, etc as iron vapour. Electrons are divided among the two gases so that each gas is charge neutral. For a plasma containing two gases, the $N-1$ species conservation equations (14) can then be replaced by a single gas conservation equation

$$\frac{\partial \rho \bar{Y}_A}{\partial t} + \nabla \cdot (\rho \bar{v} \bar{Y}_A) + \nabla \cdot \bar{J}_A = 0, \quad (18)$$

where $\bar{Y}_A$ is the sum of the mass fractions of the species in gas A and $\bar{J}_A$ is the average mass flux of the species in gas A.

The main difficulty in applying equation (18) is determining $\bar{J}_A$. Possibly the most widely used approach is the combined diffusion coefficient method, initially developed for neutral gases [66] and then for plasmas [65, 68]. If LTE can be assumed, this method is mathematically equivalent to the full multicomponent diffusion treatment for mixtures of homonuclear gases that do not react with each other. The requirement of non-reacting homonuclear gases means that a gas is equivalent to a chemical element. The diffusion mass flux of gas A can then be written

$$\bar{J}_A = \frac{n^2}{\rho} \bar{m}_A \bar{m}_B (\bar{D}_{AB}^x \nabla \bar{x}_B + \bar{D}_{AB}^P \nabla \ln P + \bar{D}_{AB}^E \vec{E}) - \bar{D}_{AB}^T \nabla \ln T, \quad (19)$$

where $\bar{m}_A$ and $\bar{m}_B$ are, respectively, the average masses of the heavy species of gas A and gas B and $\bar{x}_B$ is the sum of the mole fractions of the species of gas B. The combined diffusion coefficients $\bar{D}_{AB}^x$, $\bar{D}_{AB}^P$, $\bar{D}_{AB}^E$ and $\bar{D}_{AB}^T$ describe, respectively, diffusion due to mole fraction gradients, gradients of the total pressure, externally applied electric fields and temperature gradients. They are linear combinations of the multicomponent diffusion coefficients; expressions are given in [65]. The first three depend on only the ordinary

diffusion coefficients, while $\bar{D}_{AB}^T$ depends on both the ordinary and thermal diffusion coefficients. Values of the combined diffusion coefficients have been given in the literature for mixtures of argon and nitrogen, argon and oxygen [40], argon and helium [41], argon and hydrogen [42], air and argon, air and nitrogen, air and oxygen [69] and sulfur hexafluoride and teflon vapour [70]. Air is itself a gas mixture, and sulfur hexafluoride and teflon vapour are not homonuclear gases; however, the combined diffusion coefficient method can be applied to give approximate data for such cases.

Clearly, the combined diffusion coefficient method is a major simplification, since $N-1$ species conservation equations are replaced by one ‘gas’ conservation equation and N^2 ordinary diffusion coefficients (of which $N(N-1)/2$ are linearly independent) and N thermal diffusion coefficients (of which $N-1$ are linearly independent) are replaced by four combined diffusion coefficients (or as few as two, since $\bar{D}_{AB}^P$ and $\bar{D}_{AB}^E$ can be neglected in many cases). The method has been applied to a wide range of plasma processes; for example demixing in arcs [53, 71, 72], heat transfer from an argon–hydrogen arc to metal beads [73], arc reactors [74] and plasma torches [75] operating in gas mixtures, the entrainment of air into plasma jets [76, 77] and ablation of polymers in a circuit breaker [70, 78].

Rini *et al* [79] developed an extension of the combined diffusion coefficient method to heteronuclear gases and gases that react to each other. They did this by using stoichiometric coefficients to relate the composition of each gas to the chemical elements it contains, and thereby tracking the diffusion of each chemical element. The calculation of thermodynamic properties is implicit in the equations developed. This has the advantage that the derivatives of these properties do not have to be evaluated using computationally expensive finite difference calculations. However, it does lead to more complicated expressions for the diffusion coefficients. The method has been applied to the demixing in an inductively coupled plasma operating in air [80], and in a carbon dioxide–nitrogen mixture [81]. Orsini [82] recently developed an analytic method of calculating the derivatives of the thermodynamic properties, which could also be applied to decrease computation times. Rat *et al* [83] have extended the combined diffusion coefficient method to two-temperature plasmas, and presented values for argon–hydrogen plasmas.

Other simpler approaches have also been applied to calculate $\bar{J}_A$ in (18); these have been reviewed and compared for different scenarios [43]. In these cases,

$$\bar{J}_A = -\rho \bar{D}_{AB} \nabla \bar{Y}_A \quad (20)$$

is used, and approximate expressions used to determine the diffusion coefficient $\bar{D}_{AB}$. They include the ‘binary diffusion coefficient method’, in which $\bar{D}_{AB}$ is equated to the binary diffusion coefficient describing diffusion between two species (e.g. iron atoms and argon atoms in an argon–iron vapour plasma); this is only accurate at temperatures below about 5000 K, for which no dissociation or ionization has taken place [43]. Two approximations based on the viscosity are also used; of these, the ‘second viscosity approximation’, derived

using an expression given by Wilke [84] that interpolates the viscosity of the gases present, has been widely used for modelling diffusion of metal vapours in arcs. It is more accurate than the other approximate treatments examined, and gives results accurate to within a factor of 2 for temperatures up to about 8000 K for diffusion of vapour produced from a copper particle in an argon plasma [43].

For plasmas out of LTE, a second class of diffusion treatments has to be applied (although, as has been done for two-temperature plasmas in local chemical equilibrium [83], it is possible to derive extensions of the combined diffusion coefficient method for some specific types of non-LTE plasmas). In this second class, which is applicable to any plasma, in or out of LTE, continuity equations (14) are solved for each species, and the mass flux J_i of each species is determined using an approximate treatment, rather than the full expression (15).

The most common such approximation is to use binary diffusion coefficients, which depend on only two species, and are therefore much simpler to calculate than multicomponent diffusion coefficients, which depend on interactions between all the species in the plasma. An approach that has been applied to both plasmas and neutral gases is to use

$$J_i = -\rho D_i \nabla Y_i \quad (21)$$

for the mass flux required in (14), with the effective binary diffusivity D_i of species i given by an expression of the form

$$D_i = \frac{(1 - x_i)}{\sum_{j \neq i} x_j / \mathcal{D}_{ij}}, \quad (22)$$

where $\mathcal{D}_{ij}$ is a binary diffusion coefficient. This is often referred to as the effective binary diffusion coefficient (EBDC) approximation [85, 86]. When only two species are present, it is equivalent to (15). However, as pointed out by Ramshaw [87], when more than two species are present the approach is unphysical, since $\sum_i J_i = 0$, which is required since the diffusion velocities are defined relative to the mass-average velocity, is not satisfied. One approach is to circumvent this problem by defining J_i for one of the species as $J_j = -\sum_{i, i \neq j} J_i$. However, there is no basis for choosing which species should fill this role, and different results are obtained for different choices [87].

By using the Stefan–Maxwell equations as a starting point, it has been possible to develop approximate treatments of diffusion that utilize the binary diffusion coefficients, and that have an acceptable level of accuracy. The Stefan–Maxwell equations are

$$\sum_{j=1}^N (x_j x_i / \mathcal{D}_{ij}) (v_j - v_i) = d_i, \quad i = 1, \dots, N. \quad (23)$$

The driving force term d_i was given in (16).

Ramshaw [87] used the Stefan–Maxwell equations to derive the self-consistent effective binary diffusion coefficient (SCEBDC) approximation, so named because it removed the problem noted above with the EBDC approximation that

the J_i for the individual species do not sum to zero. In the SCEBDC approximation, a constant vector is added to the diffusion velocities, which gives an expression for J_i based on the driving force term (16) as well as the binary diffusivity. The results in Ramshaw's initial paper were only applicable to neutral gases. Ramshaw and Chang [88] extended the approach to single-temperature plasmas by including ambipolar diffusion, and subsequently to two-temperature plasmas [89] and to the general case of two-temperature magneto-hydrodynamics [90].

The SCEBDC approximation has been demonstrated to give more realistic results than the EBDC approximation. For example, Désilets *et al* [91] showed this for a single-temperature plasma in a mixture of 10 gases, benchmarking their results against the exact calculation given by equation (15). Domingo *et al* [92] considered a two-temperature nitrogen plasma jet, benchmarking their calculations against measurements. The SCEBDC method has been applied to the modelling of a range of two-temperature and non-LCE plasma processes, including low-pressure inductively coupled plasmas for semiconductor processing [93–95], nucleation in low-pressure silane–oxygen–argon plasmas [96] and atmospheric-pressure inductively coupled plasmas with departures from local chemical equilibrium [97].

Kolesnikov and Tiskii [98–100] developed generalized Stefan–Maxwell equations for any order of approximation, and further grouped species into gases, paralleling the combined diffusion coefficient method. This was important to the development of the approach of Rini *et al* [79] mentioned above. In particular, Kolesnikov [98] developed a self-consistent approach to modelling of diffusion in a two-temperature plasma that is similar to Ramshaw and Chang's work, but which takes into account higher-order approximations of the diffusion coefficients.

4.2. Influence of iron vapour in welding arcs

As an example of the importance of modelling of diffusion in plasmas, we consider the example of metal vapour diffusion in welding arcs. In TIG welding, metal evaporates from the molten weld pool. In MIG welding, metal also evaporates from the metal droplets as they form on the wire electrode and as they fall through the arc. As well as the intrinsic value of investigating the effect of the metal vapour on the arc and weld pool properties, understanding the metal vapour production and diffusion is also important since it is a precursor to the fume that forms during arc welding [101]. The hexavalent chromium fume that forms during welding of stainless steel is particularly hazardous [62].

Figure 8 shows the transport coefficients and net radiative emission coefficient of mixtures of iron vapour and argon. The presence of just 1 mol% iron vapour has a very large effect on the net radiative emission coefficient [102] and also on the electrical conductivity for temperatures up to around 10 000 K. These results are typical for metallic impurities in plasmas, since metallic species radiate more strongly, and ionize at lower temperatures, than the common plasma gases. The addition of 10 mol% iron vapour approximately doubles

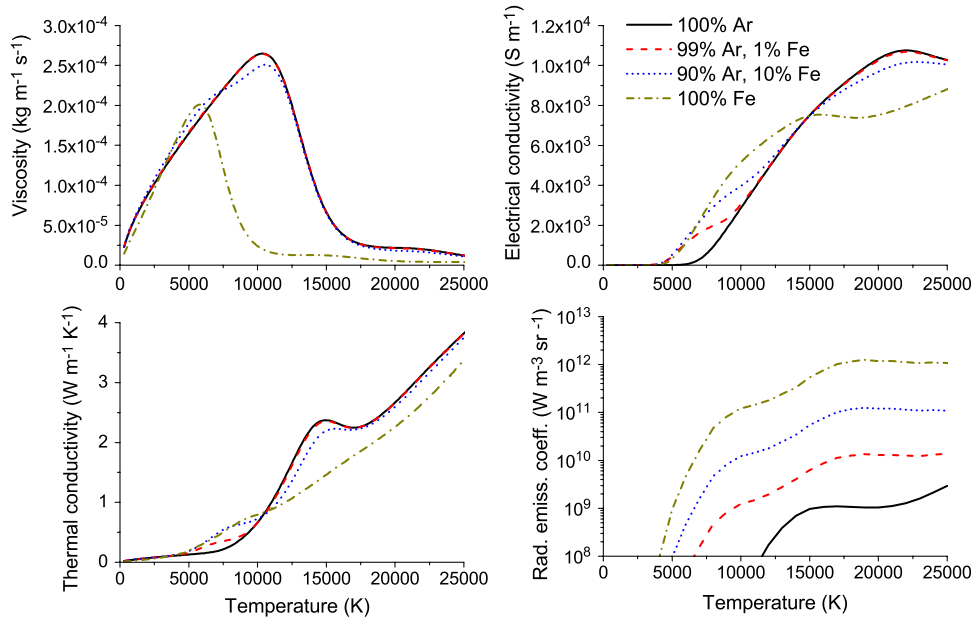


Figure 8. Dependence of viscosity, electrical conductivity, thermal conductivity and net radiative emission coefficient on temperature for argon, iron vapour, and mixtures of argon with 1 mol% and 10 mol% iron vapour, at 1 atm.

the thermal conductivity for temperatures between 5000 K and 10 000 K, while the effect on viscosity is minor even for this concentration of iron vapour.

Many researchers have modelled the production and diffusion of metal vapour in welding arcs. Typically, equation (14), written here for iron vapour, is used:

$$\frac{\partial \rho Y_{\text{Fe}}}{\partial t} + \nabla \cdot (\rho \underline{v} Y_{\text{Fe}}) + \nabla \cdot \underline{J}_{\text{Fe}} = 0. \quad (24)$$

The two main problems to be addressed are the treatment of the vaporization rate of metal, usually given by a boundary condition at the molten metal surface, and the treatment used to obtain the diffusive mass flux of metal vapour $\underline{J}_{\text{Fe}}$. Here we mention the approaches used by other researchers. In all cases, the calculations were performed in two-dimensional axisymmetric geometry, and the mass fraction of the metal vapour at the anode surface was in all but one case determined using

$$Y_{\text{m}} = \frac{P_{\text{vap}} M_{\text{m}}}{P_{\text{vap}} M_{\text{m}} + (P_{\text{atm}} - P_{\text{vap}}) M_{\text{p}}}, \quad (25)$$

where

$$P_{\text{vap}} = P_{\text{atm}} \exp \left[\frac{-H_{\text{vap}}}{R} \left(\frac{1}{T} - \frac{1}{T_{\text{b}}} \right) \right] \quad (26)$$

is the vapour pressure of the metal at the surface temperature of the anode, P_{atm} is atmospheric pressure, M_{m} and M_{p} are, respectively, the molar masses of the metal and the plasma gas, H_{vap} is the molar heat of evaporation, T_{b} is the boiling temperature of the metal at atmospheric pressure and R is the ideal gas constant. The means of calculating the surface temperature differed significantly from study to study, ranging from estimates to self-consistent calculations of heat transfer and flow in the weld pool.

Menart and Lin [103] and Zhao *et al* [104] modelled argon arcs with copper vapour present. The anode region

was not treated self-consistently; rather a surface temperature was assumed, and from this the vapour pressure of the copper vapour was determined using (26). Zhao *et al* used the binary diffusion coefficient approximation, which, as noted in section 4.1, is only accurate at temperatures for which both copper and argon are not ionized, while Menart and Lin used the somewhat more accurate second viscosity approximation, also discussed in section 4.1.

Gonzalez *et al* [105] investigated the influence of iron vapour in TIG welding arc with an iron anode. The anode surface temperature was calculated self-consistently, but flow in the liquid weld pool was not included. The binary diffusion coefficient approximation was used. Gonzalez *et al* predicted iron vapour concentrations to reach 7 mol% and 60 mol%, respectively, on axis adjacent to the anode for 200 A and 300 A argon arcs, falling to about 1 mol% and 5 mol%, respectively, 2 mm above the anode. The depth of the anode region at temperatures above the melting temperature of iron was decreased when the influence of iron vapour on the arc was taken into account.

Lago *et al* [106] modelled the influence of copper, iron and aluminium vapours in a TIG arc. They also did not model flow in the liquid weld pool, and used the vapour pressure of iron at 1000 K as a boundary condition at the anode surface. The diffusion of iron vapour was treated using the second viscosity approximation. A 200 A argon arc with no metal vapour was compared with an arc with metal vapour for the same input electrical power. The presence of iron vapour increased the voltage by 1.5 V, due to cooling of the arc by increased radiative losses, so the current was decreased commensurately. The iron vapour concentration was very high, over 40 mol%, adjacent to the anode, falling rapidly to less than 1 mol% about 1 mm above the anode. Lago *et al* found that, while the total heat flux transmitted to the anode was lower for the arc containing metal vapour, the heat flux density on axis was almost two

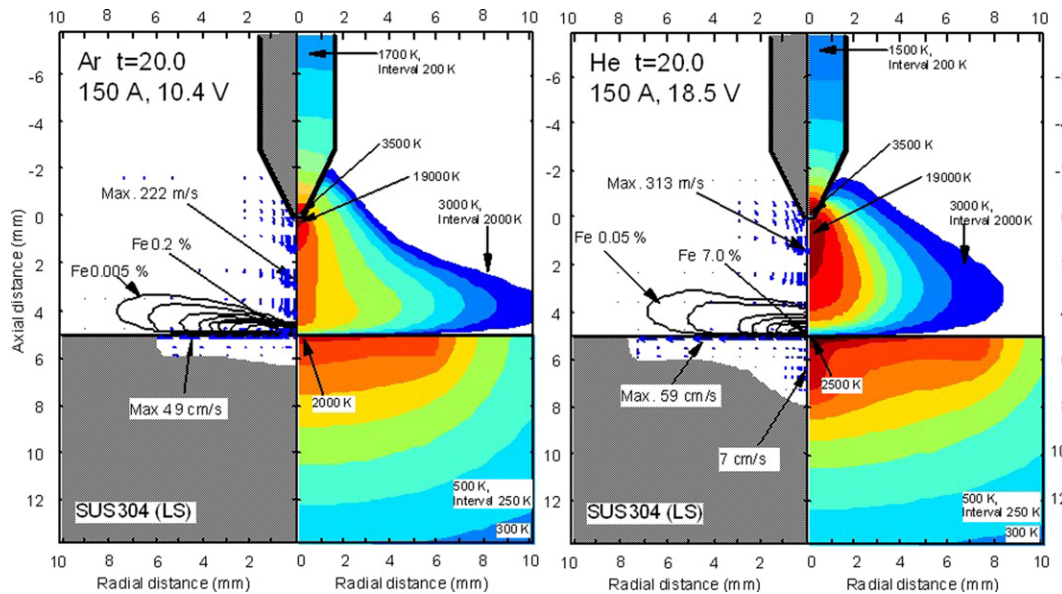


Figure 9. Temperature fields (right-hand side of each plot) and iron vapour mole fraction fields and velocity vectors (left-hand side of each plot) in the arc and electrodes for 150 A arcs in argon and helium at 1 atm, with a 304 stainless steel anode, after 20 s of operation. The maximum temperatures in the arc, cathode and anode are given. The temperature interval is 2000 K in the arc, 200 K in the cathode and 250 K in the anode. The maximum velocity is given in the arc and in the anode. The maximum values of the iron vapour concentration (in mol %) are given. The arc voltage is given for each case. After [34].

times larger. This was explained in terms of the temperature being lower at large radii, causing the electrical conductivity to be lower. The lower temperature overrode the increase in electrical conductivity at a given temperature due to the presence of metal vapour.

Tashiro *et al* [107] investigated the influence of iron vapour on a helium TIG arc. They assumed a uniform iron vapour concentration throughout the arc. They predicted a lower arc temperature due to radiative losses and changes in the electrical conductivity, and a lower heat flux density to the anode, for iron vapour concentrations from 5 mol% to 30 mol%.

Schnick *et al* [108] presented preliminary results of calculations of the influence of iron vapour in a MIG welding arc. Instead of using a boundary condition for the vapour mass fraction, an iron vapour source term was added to the right-hand side of (24), with the tip of the wire electrode assumed to vaporize at a constant rate. Schnick *et al* used the combined diffusion coefficient method to treat diffusion, and predicted a strong concentration of iron vapour near the axis of the arc.

Yamamoto *et al* [34, 35] calculated the influence of iron vapour on helium and argon TIG welding arcs. The main advance in this work was that flow in the molten weld pool was included in the model, so a more realistic surface temperature could be obtained. The second viscosity approximation was used to treat diffusion of the metal vapour in the arc; in future work, the combined diffusion coefficient treatment will be incorporated. Here we present representative results from the work of Yamamoto *et al*, and examine in detail the influence of transport properties on the boundary conditions at the arc-anode interface.

Figure 9 shows the temperatures, velocities and iron vapour mole fractions in 150 A arcs in argon and helium, for an anode composed of 304 stainless steel. The presence of

vapour from the components of the steel other than iron is not taken into account. Results are shown for a time 20 s after the initiation of the arc; the iron vapour concentration increases with time, rapidly initially, but only slowly after about 5 s, so the data shown are close to steady state [34].

The iron vapour concentration for the argon arc reaches a maximum of only 0.2 mol%, compared with 7.0 mol% for the helium arc. This is because the weld pool temperature is higher for the helium arc, as a consequence of the greater heat flux density (see figure 7(a)). As can be seen from (26), the vapour pressure increases rapidly as the temperature approaches the boiling point (about 2980 K for 304 stainless steel [109]), so the 500 K temperature difference makes a large difference to the vapour pressure.

As a consequence, the influence of iron vapour is much larger for the helium arc. Figure 10 shows the transport coefficients and radiative emission coefficient for mixtures of helium and iron vapour. As was the case for argon (figure 8), the most dramatic effects of the addition of iron vapour are found in the electrical conductivity and net radiative emission coefficient. The electrical conductivity is increased over a greater temperature range for helium: up to 14 000 K, compared with about 10 000 K for argon, for 1 mol% iron vapour. The addition of 10 mol% iron vapour significantly decreases the thermal conductivity for temperatures above 15 000 K; however, it can be seen in figure 9 that there is only a small region of the arc for which the temperature is above 15 000 K and the iron vapour concentration is less than 1 mol%, so this change will have little effect on the arc properties. Finally, the addition of 10 mol% iron vapour also increases the viscosity for temperatures below about 7000 K; again, these parameters only occur in a very small region near the anode.

The results presented in section 3 indicated that the product of the enthalpy and density was critical in determining

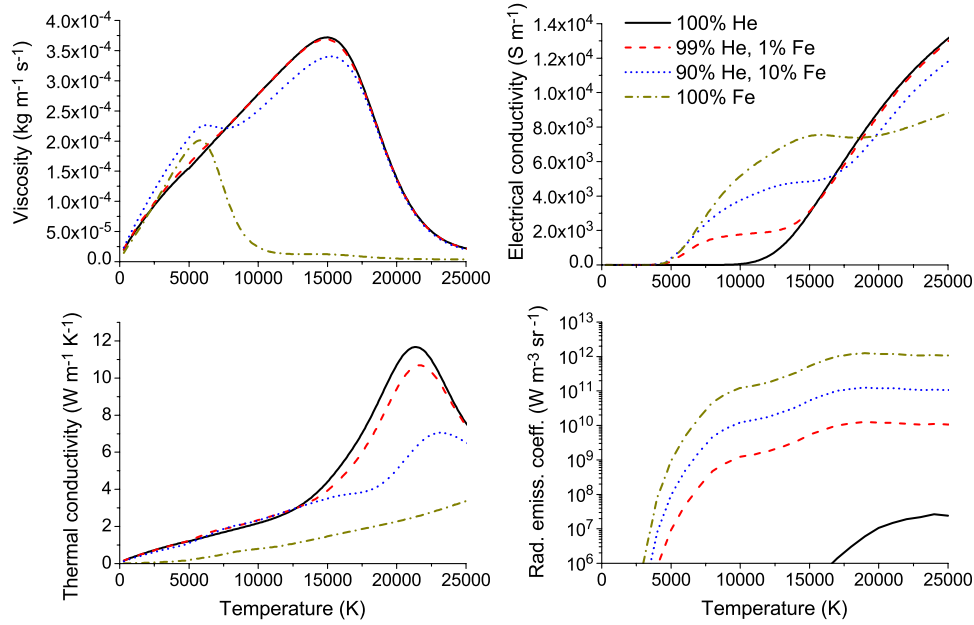


Figure 10. Dependence of viscosity, electrical conductivity, thermal conductivity and net radiative emission coefficient on temperature for helium, iron vapour, and mixtures of helium with 1 mol% and 10 mol% iron vapour, at 1 atm.

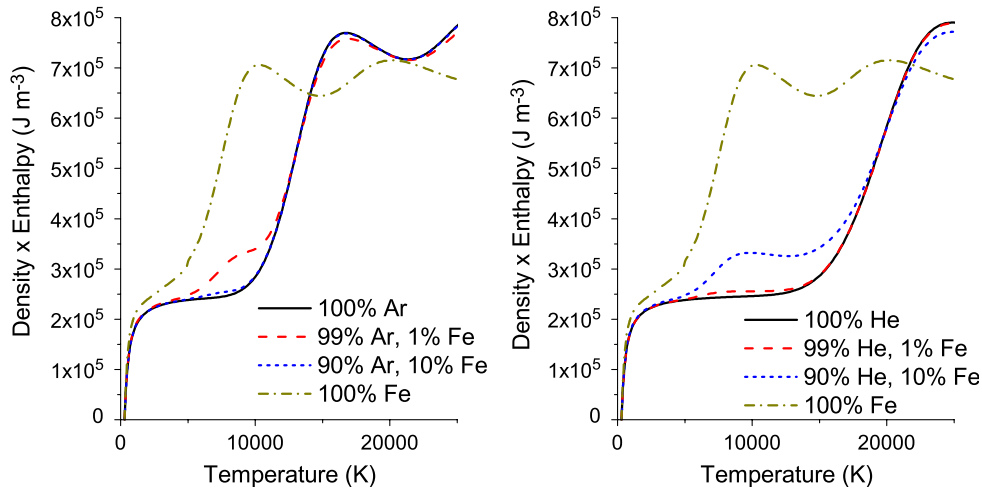


Figure 11. The product of enthalpy and mass density of (a) argon, iron vapour and mixtures of argon with 1 and 10 mol% iron vapour and (b) helium, iron vapour and mixtures of helium with 1 and 10 mol% iron vapour, at 1 atm.

arc properties. Figure 11 shows the influence of adding iron vapour on this product for argon and helium plasmas. The effect is insignificant for 1 mol% iron vapour. However, adding 10 mol% iron vapour increases the product of enthalpy and density by up to about 30% for argon for temperatures between 5000 and 10 000 K, and by a similar percentage for helium for temperatures between 5000 and 18 000 K.

Figure 12 shows the radial dependence of the heat flux density, current density, shear stress and arc pressure at the anode for argon and helium arcs, ignoring and including the influence of iron vapour. The presence of metal vapour decreases the heat flux density and current density on axis. The effect is much greater for the helium arc since the iron vapour concentration is much larger. The change is mainly because of the higher electrical conductivity at low temperatures, which facilitates conduction in the cooler regions at greater radii, thus leading to a less peaked current density profile near the

anode. This in turn influences the heat flux density through the electron capture term in (9). The presence of iron vapour is also expected to have an effect on the helium arc through the increased product of enthalpy and density. However, the influence on the temperature distribution is only minor, since the iron vapour concentration is small except close to the anode. We conclude that the presence of metal vapour will decrease the weld pool depth, through both the decreased heat flux to the weld pool and the decreased magnetic pinch force in the weld pool.

Comparison of figures 2 and 9 shows that the presence of metal vapour also decreases the helium arc temperature near the anode. This is partly because of the lower current density near the axis, which leads to weaker ohmic heating in this region, and partly because of the increased radiative emission, which cools the arc. We found that the presence of metal vapour leads to a decrease in the arc voltage. The

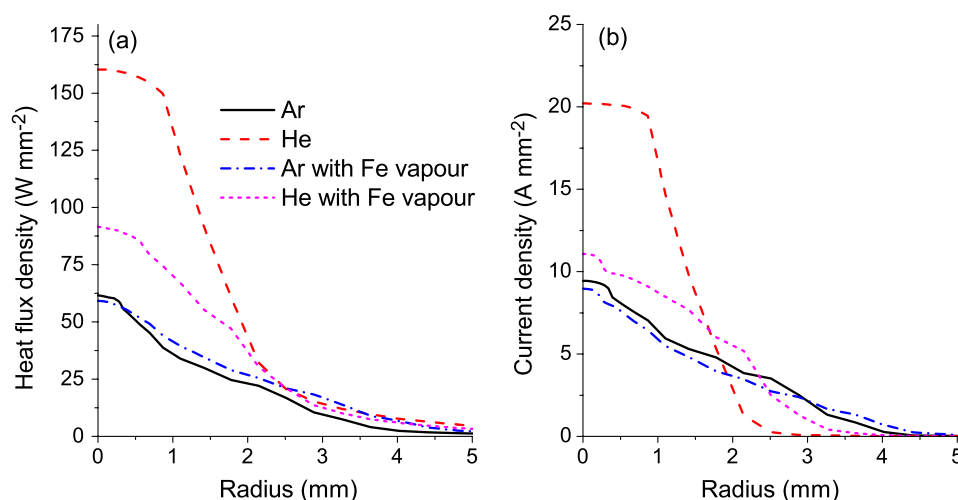


Figure 12. Radial dependence of (a) heat flux density and (b) current density, both at the anode, for arcs in argon and helium with 304 stainless steel anode, both neglecting and including the influence of metal vapour.

higher electrical conductivity is expected to decrease the voltage, while the increased radiative emission will increase the voltage (by cooling the arc, therefore decreasing the electrical conductivity). For the relatively low concentrations of metal vapour found here, the former effect is dominant. It is interesting to note that when larger concentrations of vapour are present, the latter effect can dominate, as seen for example by Lago *et al* [106], who found a voltage increase of 1.5 V for a 200 A arc in argon with iron vapour concentration as high as 40 mol%, and Gonzalez *et al* [105], who found a 0.6 V increase for a 200 A arc, and a 4.5 V increase for a 300 A arc in argon, for iron vapour concentrations peaking at 7 mol% and 60 mol%, respectively. Tashiro *et al* [107] found that, for uniform iron vapour concentrations of 5, 10 and 20 mol% in a helium arc, the voltage was lower than for a pure helium arc, while for a uniform iron vapour concentration of 30 mol%, the voltage was higher.

The results presented here are not directly comparable to those of most of the other authors, since the arc current used is lower (150 A compared with at least 200 A). This will lead to lower anode temperatures and a lower concentration of iron vapour. However, even taking this into account, the iron vapour concentration found for an argon arc (0.2 mol%) is significantly lower than that predicted by Gonzalez *et al* [105] (7 mol% in a 200 A arc) and Lago *et al* [106] (40 mol% in a 200 A arc). The prediction by Lago *et al* that the energy flux density on axis is larger for the arc with metal vapour is also not borne out by our results. It is expected that our results should be more reliable because of the self-consistent calculation of anode surface temperature that takes into account flow in the weld pool.

5. Conclusions

We have reviewed the methods used to model thermal arc plasmas, both in pure gases and in mixtures of gases. Our focus has been on the influence of thermophysical properties on the parameters of TIG welding arcs, particularly those that

affect the weld pool. For mixtures of gases, we have assessed the different treatments of diffusion that have been applied.

We have illustrated the influence of thermophysical properties using two studies. The first study investigated the effect of changing each of six different thermophysical properties on the properties of an argon TIG welding arc. The most important properties were found to be the mass density and specific heat (or enthalpy); increasing the product of these properties leads to a constriction of the arc via the ‘thermal pinch effect’. Decreasing the electrical conductivity also leads to some constriction since it forces the current to flow in the hotter central regions. Contrary to some reports in the literature, increasing the thermal conductivity gives a less constricted arc over most of its length, although the heat flux to the anode was nevertheless increased. The main influence of the viscosity was on the flow velocities in the arc, which affected the arc pressure and shear stress at the anode. For the 150 A arcs considered here, altering the radiative emission coefficient had little influence; for higher current arcs, it is expected to be more important. By examining the influence of the different properties on the heat flux, current density and shear stress at the anode, it was concluded that the weld pool depth would be increased by using shielding gases with high specific heat, thermal conductivity and viscosity.

The second study considered the effect of metal vapour on welding arcs. The concentrations of metal vapour predicted for 150 A arcs were very low (up to 0.2 mol%) for argon shielding gas, but higher (up to 7 mol%) for helium shielding gas. The metal vapour affects the arc mainly by increasing the electrical conductivity at lower temperatures, and increasing the radiative emission coefficient. For the concentrations reported here, the former effect was more important, and led to lower heat flux density and current density at the weld pool surface, implying a shallower weld pool.

The results presented underlined the importance of the thermophysical properties in determining the characteristics of welding arcs, including the weld pool depth and geometry. Increased understanding of the influence of these properties will continue to allow improved shielding gases and welding

processes to be developed. Although we have concentrated on arc welding, it is clear from the results presented that the plasma gas properties will be of great importance in any thermal plasma process.

The models we have presented have many of the features required to achieve a realistic model of arc welding. These include the inclusion of the electrodes, arc and weld pool in the computational domain in a self-consistent manner, accurate calculation of the thermophysical properties of the shielding gas and the consideration of the metal vapour flux from the molten weld pool. There are, however, at least two further features that will be necessary before a fully predictive model can be realized. These are inclusion of the motion of the electrode and arc relative to the workpiece, and treatment of deformation of the weld pool surface. The first of these requires a fully three-dimensional model [110], while the second requires the use of specialized methods such as the volume-of-fluid method [111] to track the weld pool surface. To model MIG welding reliably, the effects of the formation, transport and vaporization of the droplets formed on the wire electrode have also to be taken into account [112]. Finally, more realistic welding geometries than the flat-plate geometry considered here will require three-dimensional treatments. While each of the above issues has been successfully tackled, we are not aware of any model that contains all the features, so the goal of a fully predictive model of arc welding is yet to be fully realized.

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